

Functional and Non-Functional Requirements

Functional Requirements

A. Greenhouse/Garden Area

Plant Care

FR1: The system must use different plant care strategies (watering, fertilizing, pruning, misting) based on plant type.

- **Design Pattern:** Strategy Pattern

FR2: The system must attach and switch care strategies at runtime for individual plants.

- **Design Pattern:** Strategy Pattern

FR3: The system must invoke care strategies through command objects.

- **Design Pattern:** Command Pattern

Plant Life Cycle

FR4: The system must manage plant lifecycle states (Planted, InNursery, ReadyForSale, Withering).

- **Design Pattern:** State Pattern

FR5: The system must advance plants through lifecycle states based on care actions or neglect.

- **Design Pattern:** State Pattern

FR6: The system must trigger automatic withering when no staff can respond to care needs.

- **Design Pattern:** State Pattern

Inventory

FR7: The system must notify observers when stock quantity changes or plants reach maturity.

- **Design Pattern:** Observer Pattern

FR8: The system must provide centralized global access to inventory management.

- **Design Pattern:** Singleton Pattern

Stock

FR9: The system must instantiate plants with category profiles (Flower, Tree, Succulent) exposing species data.

- **Design Pattern:** Bridge Pattern

FR10: The system must create different plant types through a unified factory interface.

- **Design Pattern:** Abstract Factory Pattern //gone?

FR11: The system must provide configurable construction of composite plant structures with uniform treatment.

- **Design Pattern:** Builder Pattern + Composite Pattern

B. The Staff

Plant Care

FR12: The system must define a maintenance template allowing staff subclasses to customize specific steps.

- **Design Pattern:** Template Method Pattern //gone?

FR13: The system must encapsulate staff actions into command objects that can be executed and undone.

- **Design Pattern:** Command Pattern

FR14: The system must register, clone (via prototype), inject target plants, and delete commands after execution.

- **Design Pattern:** Command Pattern + Prototype Pattern //gone?

Plant Life Cycles & Inventory

FR15: The system must coordinate staff-plant communication without direct dependencies.

- **Design Pattern:** Mediator Pattern

FR16: The system must notify staff observers on plant state changes and build appropriate care commands.

- **Design Pattern:** Observer Pattern + Command Pattern

Customer Interaction

FR17: The system must delegate customer requests through a staff chain based on authority level.

- **Design Pattern:** Chain of Responsibility Pattern

FR18: The system must dispatch commands to appropriate staff chains and handle graceful failures.

- **Design Pattern:** Chain of Responsibility Pattern

C. The Customers and The Sales Floor

Customer Browsing

FR19: The system must allow customers to browse plants through a unified interface.

- **Design Pattern:** Facade Pattern

Personalization

FR20: The system must allow customers to personalize orders through a customizable building process.

- **Design Pattern:** Builder Pattern

FR21: The system must allow customers to dynamically add enhancements to plants.

- **Design Pattern:** Decorator Pattern

Interaction with Staff

FR22: The system must route customer requests to appropriate staff through a chain of handlers.

- **Design Pattern:** Chain of Responsibility Pattern

FR23: The system must coordinate customer-staff interactions without direct dependencies.

- **Design Pattern:** Mediator Pattern

Sales and Transactions

FR24: The system must organize plants into composite structures for individual or bundled sales.

- **Design Pattern:** Composite Pattern

FR25: The system must record and restore previous transaction states.

- **Design Pattern:** Memento Pattern

FR26: The system must adapt external payment methods to internal payment processing.

- **Design Pattern:** Adapter Pattern

FR27: The system must update inventory and dispatch purchase commands when transactions complete.

- **Design Pattern:** Observer Pattern + Command Pattern
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Non-Functional Requirements

Extensibility

NFR1: The system must support adding new species, strategies, commands, or staff roles without modifying existing client.

Scalability

NFR2: The system design must scale to accommodate growing numbers of plants, staff members, and care strategies.

Coupling & Modularity

NFR3: The system must maintain loose coupling through bridge, strategy, and command patterns to isolate subsystems and prevent ripple effects from changes.

Runtime Flexibility

NFR4: The system must select and clone strategies and commands at runtime, enabling dynamic behaviour shifts without recompilation.